Danial Nahofteh

Computer Engineering ◇ Tehran, Iran

PERSONAL DETAILS

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RESEARCH INTERESTS

Cloud, Edge Computing ◇ Distributed Systems ◇ Blockchain ◇ Artificial Intelligence

EDUCATION

Amir Kabir University of Technology (Tehran Polytechnic), Iran *September 2024 - Present*
Master's degree in Software Engineering.

- GPA: 16/20 (3.2/4)
- Master's Thesis: Cold Start Mitigation through Energy Aware Serverless Functions Placement.
- Supervisor: Dr. Amir Kalbasi.
- Selected Courses: Cloud Computing (18.6/20), Distributed Systems (17/20), Performance Evaluation of Computer Systems (17/20).

Islamic Azad University of Mashhad, Iran *September 2019 - September 2023*
Bachelor in Computer Engineering.

- GPA: 18.09/20 (3.62/4)
- Top 10% students of the University (based on GPA 3.62/4.0 (18.09/20)).
- Selected Courses (grade 4/4): Fundamentals of Computer and Programming, Systems Analysis and Design, Computational Intelligence, Fundamentals of Speech and Language Possessing.

RESEARCH EXPERIENCE

Cold Start Mitigation through Energy Aware Serverless Functions Placement
Master Thesis (AUT). *June 2025 - December 2026 (expected)*
Supervisor: Dr. Amir Kalbasi

- Conducting research on optimizing placement of serverless functions (FaaS) to reduce cold-start latency and energy consumption in cloud environments.
- Developing an energy-aware scheduling and resource allocation model using simulation tools such as CloudSim and real-world cold-start traces.
- Aiming to contribute to sustainable and high-performance cloud computing by balancing latency reduction and energy efficiency.

Sustainable Cloud Computing Through Task Assignment
Seminar report (AUT). *September 2024 - September 2025*
Supervisor: Dr. Amir Kalbasi, Course Teacher: Dr. Mohammad Reza Meybodi

- Investigated energy-efficient and carbon-aware task scheduling strategies for sustainable cloud computing.
- Reviewed multi-objective optimization techniques, including heuristic, metaheuristic, and AI-based task assignment algorithms.

PUBLICATIONS

SpeQ: Speech Quality Estimation in MoQ *Submitted (under review)*
M. Esfandiari Sabzevar, **D. Nahofteh**, B. Barghbani, M. Hoi, M. Dolati, F. Tashtarian.
Mile High Video Conference (MHV).

- Studied subscriber-side speech-quality estimation in Media over QUIC (MoQ) as a signal for adaptive Opus bitrate selection in real-time voice streaming.

WORK EXPERIENCE

Backend and Blockchain Developer
kifpool.me. *September 2024 - Now*

- Developed web services using Golang and a microservices-based architecture.
- Designed and implemented blockchain-based backend services for a cryptocurrency exchange platform.
- Built a high-performance matching engine for P2P user order matching.

Backend Developer
Nasim Telecom. *November 2021 - September 2024*

- Designed and implemented a distributed web application using Golang, Microservices architecture, gRPC, and GraphQL.
- Developed a distributed call center software cluster leveraging ETCD, Python, and Golang.
- Built and configured a Kubernetes cluster for a cloud XaaS platform using Docker.

TEACHING EXPERIENCE

Teacher Assistant

- Teaching python programing and Django(python web framework) and basic web Design to students.

Advanced Programming

- Teaching freshman students programming topics, including the STL library of C++, Time Complexity, and Introduction to Programming Contests.

SKILLS

Programming Languages	Golang, Python, Rust, Solidity, Java, C++, C, SQL, Matlab
Software Libraries	scikit-learn, Numpy, Pandas, Keras, Matplotlib
Frameworks	Pytorch, TensorFlow, Django, DjangoRest, Gin, React
Databases	MongoDB, Etcd, Redis, Neo4j, MySql, SqlServer
Simulation Tools	CloudSim, SimGrid
Operating systems	Linux, Windows, MacOS
Other Related Skills	AWS, GCP, Terraform, Kubernetes, Docker, Vault, OpenWhisk, Git, gRpc, proto3, GraphQL, Microservices, Clean Coding, Nats

SELECTED PROJECTS

Cloud Ticket Reservation System GitHub

- Developed a scalable, serverless event ticketing system on AWS using Lambda, DynamoDB, SQS, and S3 with real-time seat reservation and queue-based purchase handling.

Cloud Live Migration (Google Cloud to AWS)

- Researched, planned, and executed a full live migration of services and data from Google Cloud Platform (GCP) to Amazon Web Services (AWS), ensuring minimal downtime and full workload compatibility.

Real Estate Management Backend System — National Elites Foundation

- Designed and developed the backend of a comprehensive property management and monitoring platform for the National Elites Foundation, enabling secure data handling and efficient asset oversight.

LANGUAGES

Persian	Native
English	Advanced IELTS Score:6.5 (Reading:6.5, Listening:6, Speaking:6.5, Writing:6)

HONORS AND AWARDS

2024	Ranked 94th nationwide among participants in the Iranian M.Sc. entrance exam in Computer Engineering.
2024	Certificates in Python, Pandas, Data Cleaning, and Data Visualization available here.
2021	Ranked 5th, 2021 TCPC competitions of Tabriz University.
2021	Ranked 42th, 2021 ACPC competitions of AmirKabir University of Tehran.
2021	Selected top lecturer, in webinars organized in 2021 at Azad university of Mashhad.
2020	Ranked 39th, 2020 ICPC Asia Tehran Regional Contest certificate available at here.
2020	Ranked 2nd, 2020 Internal competitions of Azad University of Mashhad.